Cheatsheet AZ-305 — 15 · Data platforms, analytics and integration

1. Data services map

Service	Category	What for	Signal
Azure Data Factory (ADF)	Integration / ETL-ELT	Low-code pipelines, 90+ connectors, mapping data flows (code-free Spark), Self-hosted IR (on-prem), Azure-SSIS IR	"batch ETL", "low-code", "on-prem sources", "SSIS"
Synapse pipelines	Integration	ADF built into Synapse	Same as ADF inside Synapse
Synapse Analytics	Unified analytics platform	Dedicated SQL pools (MPP warehouse), serverless SQL (query the lake), Spark pools, pipelines, Synapse Link	"data warehouse", "query the lake directly"
Microsoft Fabric	Unified analytics SaaS	Lakehouse, Warehouse, Data Factory, Real-Time, Power BI with OneLake and **Direct Lake**	"single SaaS platform", "Power BI without import"
Azure Databricks	Managed Spark	Data engineering, ML, Delta Lake; Unity Catalog / Credential Passthrough	"Spark", "notebooks", "ML at scale"
Azure Stream Analytics	Real-time streaming	**SQL** queries over streams (Event Hubs, IoT Hub, Blob) with time windows; outputs to SQL, Cosmos, Power BI, Functions…	"real-time", "windowing", "SQL-like over streams", "Power BI dashboard live"
Azure Data Explorer (ADX / Kusto)	Log and time-series analytics	Massive ingestion, interactive **KQL** queries over TB in seconds; foundation of Log Analytics and Sentinel	"telemetry analytics", "time series", "interactive ad-hoc over billions of rows"
Event Hubs / IoT Hub	Ingestion	See below	
Data Lake Storage Gen2	Lake storage	Hierarchical namespace, POSIX ACLs, Hadoop	"lake", "Hadoop", "directories by date"
Cosmos DB	Global NoSQL	See cheatsheet 09	
Azure SQL / Hyperscale	Relational	See cheatsheet 09	
Azure Cache for Redis	Cache	Session state, read cache	"near the app"
Microsoft Purview	Data governance	Catalog, classification, lineage, glossary	"discover and classify sensitive data"
Azure Data Share	Sharing	Snapshots to other organizations	"partner", "restricted access to snapshots"
Azure AI Search	Search	Full-text, vector indexes	"search across documents"

2. Data Factory: pieces that come up

Piece	Detail
Pipeline / activities	Copy, Data Flow, Lookup, ForEach, Execute SSIS, Databricks notebook…
Integration Runtime (IR)	**Azure IR** (cloud, managed) · **Self-hosted IR** (installed on-prem/VM for private sources; formerly Data Management Gateway; HA with multiple nodes) · **Azure-SSIS IR** (runs SSIS packages; SSISDB in Azure SQL DB or MI)
Mapping data flows	Visual transformations executed on managed Spark
Triggers	Schedule · Tumbling window (non-overlapping windows, retries, dependencies) · Event-based (Blob created/deleted via Event Grid) · Manual
Linked services / datasets	Connections and structures
Managed VNet + Private Endpoints	Azure IR inside a managed network to reach private PaaS
CI/CD	Git integration (Azure Repos / GitHub), ARM export
Monitoring	Azure Monitor, alerts on pipeline failure

- ADF **moves and transforms**; it does **not** share data with third parties (Data Share) nor analyze in real time (Stream Analytics).

3. Streaming: who does what

Stage	Service
Device telemetry ingestion (bidirectional, device management, DPS)	**IoT Hub**
App event / log ingestion (unidirectional, high volume, Kafka)	**Event Hubs**
Automatic capture to Storage/ADLS by date	**Event Hubs Capture**
Real-time processing with SQL and windows	**Stream Analytics**
Real-time processing with code / Spark	**Functions** (per event) / **Databricks / Synapse Spark** Structured Streaming
Interactive analytics over ingested data	**Data Explorer**
Live dashboard	**Power BI** (streaming dataset from Stream Analytics)
Store results	SQL, Cosmos DB, ADLS
Edge	**IoT Edge** (modules on the device; Stream Analytics on IoT Edge)

- "IoT devices" + "device provisioning / twins / commands to devices" → **IoT Hub**; only massive inbound telemetry without management → **Event Hubs** is cheaper.
- "hot path" (seconds) → Stream Analytics; "cold path" (batch) → ADF/Synapse; "warm path" (minutes, interactive) → Data Explorer.

4. Warehousing and lakehouse: choosing the engine

Need	Engine
Relational MPP warehouse, few heavy queries	Synapse **dedicated SQL pool** (≤128 concurrent)
Relational warehouse, **high concurrency (>128)**, up to 100 TB	**Azure SQL Hyperscale**
Query lake files without loading (pay per TB scanned)	Synapse **serverless SQL**
Spark transformations, Delta Lake, ML	Databricks / Synapse Spark / Fabric
Analytics over Cosmos DB without ETL	**Synapse Link** (analytical store)
Analytics over Azure SQL without ETL	Synapse Link for SQL
Logs / time series at scale	Data Explorer
Power BI over the lake without import	Fabric **Direct Lake**
Saturated Fabric capacity	Scale the SKU (F4 → F8…), optimize refreshes, workspace governance

5. Purview (data governance)

Feature	What it does
Data Map	Scans sources (Azure, on-prem via self-hosted IR, AWS, Power BI) and builds the map
Data Catalog	Asset search, business glossary, owners
Classification	Detects sensitive data (PII, cards) with built-in and custom rules
Lineage	Source → transformations (ADF, Synapse) → reports
Data Estate Insights	Coverage and sensitivity metrics
Signal	"catalog", "classify sensitive data across the estate", "lineage", "data governance"

- Purview ≠ Azure Policy (governs resources, not data) ≠ Defender (security) ≠ Data Share.

6. Databricks: exam points

Topic	Rule
User identity down to ADLS (per-folder ACLs)	**Credential Passthrough** (Personal/Single User compute) or **Unity Catalog** with external locations
On-prem apps calling the workspace, staff turnover	**Service principal** (not PAT, not managed identity outside Azure)
Secrets	Secret scopes backed by Key Vault
Query performance	Photon (engine), not security
Private networking	VNet injection, Private Link
Tiers	Standard vs Premium (granular RBAC, credential passthrough, audit logs)

7. Decision tree

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What do I do with the data?
├─ Move / transform in batch, low-code ..... Data Factory (Self-hosted IR if on-prem)
├─ Process a real-time stream with SQL ..... Stream Analytics
├─ Process a stream with code ..... Functions / Spark streaming
├─ Ingest from devices with management ..... IoT Hub
├─ Ingest massive app events ..... Event Hubs
├─ Query logs/telemetry interactively ..... Data Explorer
├─ Relational MPP warehouse ..... Synapse dedicated pool (Hyperscale if >128 concurrent)
├─ Query the lake without loading ..... Synapse serverless
├─ Spark / ML ..... Databricks / Synapse Spark
├─ Analytics over Cosmos DB without ETL ..... Synapse Link
├─ Catalog and classify data ..... Purview
├─ Share with a partner ..... Data Share
└─ Single SaaS platform with Power BI ..... Fabric
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8. Signals → answer

Wording	Answer
"low-code batch ETL", "mapping data flows"	Data Factory
"run existing SSIS packages"	Azure-SSIS IR
"ADF must read SQL Server on-prem"	Self-hosted Integration Runtime
"real-time aggregation with time windows, SQL syntax"	Stream Analytics
"device twins, provisioning, cloud-to-device"	IoT Hub
"interactive queries over billions of log rows"	Data Explorer
"query JSON files in the lake directly"	Synapse serverless SQL
"50 TB relational + 300 concurrent reads"	Hyperscale
"near real-time analytics on Cosmos DB"	Synapse Link

Wording	Answer
"user identity enforces folder ACLs in Databricks"	Credential Passthrough
"classify sensitive data across sources, lineage"	Purview
"give Fabrikam restricted access to snapshots"	Data Share
"reduce table scans in Table Storage"	Migrate to Cosmos DB with a suitable partition key

9. Mistakes to avoid

1. Stream Analytics for batch ETL → ADF.
2. Event Hubs when device management is required → IoT Hub.
3. Synapse dedicated pool with high concurrency → Hyperscale.
4. Purview to govern resources → Policy.
5. Data Factory to share with third parties → Data Share.
6. PAT in Databricks with staff turnover → service principal.